

Two Selectivities for One Decision

Spatial filtering of uncued distractors and a value gate in a recurrent attention agent

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Abstract

An agent that must act on changes at a cued location while a competing, irrelevant change flickers elsewhere faces two problems at once: a spatial problem — respond to the cued side, not the distracting one — and an economic problem — some trials are worth more than others. We dissect, mechanistically, how a compact recurrent vision agent trained only by reinforcement solves both. The agent’s perceptual core is a pair of single-head cross-attention streams that share one recurrent memory but are read out separately: a grounded priority stream drives the policy and a memory-gated value stream drives the value estimate. Using a reconstruction of its attention verified to reproduce the trained network to numerical precision, we map where the agent looks, decode what each stream holds, and causally perturb the real circuit. Four findings organise the agent’s behaviour. (1) A causal division of labour. Biasing the priority stream’s attention swings the decision from a 1% to a 100% hit rate while leaving the critic’s value untouched; biasing the value stream never moves the decision and only shifts value and uncertainty — the policy and the valuation are causally separate, each controlled by the stream that feeds it. (2) The distractor is filtered in the decision pathway, not globally. The priority stream concentrates on the cued quadrant (0.26) and holds the distractor near zero (0.004), and a distractor onset moves its attention there by only +0.004; the value stream, by contrast, does read the distractor, so the competing change informs value but can never reach the response — and across thousands of trials not one false alarm follows a distractor. Forcing priority attention back onto the distractor manufactures the very false alarms the filter prevents (d' 5.5 $\rightarrow$ -4.9), so the suppression is active and causal. (3) Value reshapes the filter. On high-reward trials the priority filter is sharp and target-locked (0.33) with the distractor cold (0.005); on low-reward trials it is diffuse and leaks onto the distractor (0.13) — the agent invests selectivity where it pays. (4) A value gate over a near-perfect detector. On engaged (red, green) trials the agent is an excellent, distractor-immune detector (hit $\approx$ 0.83, false alarms $<$ 0.01, $d' \approx$ 3.7); on low-value (blue) trials it abandons the task, bailing at the second

frame — driven by a calibrated critic that prices the reward order precisely (4.4:2.6:0.2) and values low-reward detection at essentially zero. The agent’s modest aggregate accuracy (0.60) is not a failure of distraction control, which is near-perfect, but a deliberate refusal to work for little. Where the agent looks solves the distractor; what the trial is worth decides whether — and how sharply — it looks at all.

1. Introduction

Selective attention is usually framed as a spatial problem: a cue indicates where something relevant is likely to occur, and the task is to favour that location over others. Real environments add a second demand. Competing events occur at the unattended locations — distractors that share the target’s features and timing but carry no reward — and the agent must actively not act on them. A change-detector that fires whenever anything changes is useless in clutter; the value of selective attention is precisely that it lets a system ignore salient-but-irrelevant events. Layered on top is an economic demand: when outcomes differ in value, an agent should spend its limited effort where the payoff justifies it.

We ask how a learned recurrent agent meets both demands, and crucially where in its computation each is met. The agent studied here was trained purely by reinforcement on a cued change-detection task in which:

- a coloured cue marks one side of the display; the rewarded target change, when it occurs, appears on that side (at the cued quadrant with probability equal to the cue’s displayed reliability, otherwise at the side’s other quadrant);
- on half of trials an uncued distractor change appears on the opposite side, with an independent onset time and magnitude, carrying no reward — pressing for it is a false alarm;
- the cue’s colour sets the reward at stake on a correct detection (red = 5, green = 3, blue = 1).

The agent must detect a cued-side change, suppress an uncued-side distractor, and weigh how much each trial is worth — all from delayed reward, with no supervision of where to look, what to ignore, or when to act. Its architecture is a dual-stream recurrent attention encoder whose two streams share a single recurrent memory and are read out separately by the policy and value heads; on the distractor-free version of the task this same architecture recovers the hallmark behavioural signatures of spatial cueing. Here we put it under distraction and reward pressure and take the circuit apart.

The account we develop has two organising ideas. First, the agent’s two streams perform a causal division of labour — the priority stream carries the decision and the value stream carries the valuation — that we establish directly by perturbing each and watching what moves. Second, on this substrate the agent runs two selectivities: a spatial filter that, sitting specifically in the decision pathway, removes the distractor from the policy’s read while the value pathway still tracks it; and a value gate, driven by the critic, that decides whether the trial is worth engaging — and, we find, even tunes how sharply the spatial filter is deployed. The first selectivity is what makes the agent robust to distraction; the second is what makes its aggregate performance look modest while its engaged performance is near-perfect. We proceed in eight steps — behaviour, psychophysics, priority attention, value attention, value-modulated attention, representational decoding, causal perturbation, and the value function — each resting on a reconstruction of the agent’s attention verified against the trained network to numerical precision, so that the causal manipulations act on the real circuit.

1.1 Background and related work

The paradigm is a reinforcement-learning rendering of cued change detection with distractors, a staple of the covert-attention literature, where the signatures of interest are improved sensitivity at validly cued locations, resistance to salient distractors, and reward-sensitive engagement. Architecturally the agent is a recurrent attention network: a transformer-style cross-attention wrapped in a recurrence so that its keys and values are the network’s own evolving memory, read out by an actor and a distributional critic. The questions here — whether and how the agent suppresses an uncued distractor, where in the circuit the suppression lives, and how it trades detection against value — concern the strategy a reward-trained agent discovers when the task demands both spatial and economic selectivity, and the mechanism that implements it.

2. Methods

2.1 The cued change-detection task with distractors

Each trial is a 29-step movie of 50×50 RGB frames. A coloured cue marks the left or right side; the cue’s completeness encodes its reliability (proportion $\in \{0.25, 0.5, 0.75, 1.0\}$) and its colour encodes the reward on a correct hit (red = 5, green = 3, blue = 1). Four oriented Gabor patches are shown, one per quadrant (quadrants 0, 1 = left side; 2, 3 = right side), jittered by orientation noise each frame. On half of trials a target change occurs at a time drawn uniformly from $t \in [11, 25]$: it lands on the cued side, at that side’s primary (cued) quadrant with probability equal to the displayed reliability and at the side’s secondary quadrant otherwise. Independently, on half of trials a distractor change occurs on the uncued side — a uniformly chosen quadrant, with its own onset time and its own orientation step — carrying no reward. The agent emits wait or press each step. Pressing before the target change ends the trial with zero reward (a false alarm; this is where a press driven by an earlier distractor lands); pressing at or after the target change on a target trial yields the cue-colour reward; correctly withholding through a no-target trial also yields it. Reaction time is the number of frames between target change and press, defined on hits.

2.2 Architecture

Each frame is cut into a 10×10 grid of 5×5 patches and embedded into $N = 100$ tokens of width $d = 128$. Writing X for the patch tokens and H_1, H_2 for two per-token recurrent memories carried from the previous frame, the dual-stream encoder computes, with a single attention head each,

$$Z_{\text{pri}} = X + \text{attn}(Q=X, K=V=[H_1 \parallel H_2]) + \text{FFN}(\cdot), \quad (\text{priority; residual } X)$$

$$Z_{\text{val}} = H_2 + \text{attn}(Q=X, K=V=H_2) + \text{FFN}(\cdot), \quad (\text{value; residual } H_2).$$

The priority stream keeps the raw image on its residual and attends from the image into both memories (a source-tagged concatenation, $2N$ keys, each memory’s block tagged so the attention can distinguish the fast sensory trace from the slow deep state); the value stream keeps the deep memory on its residual, so the current image can only re-gate the stored read (N keys). Each attention block adds a learned positional embedding to the memory keys before the softmax and applies a two-layer feed-forward transform with a residual after the attention. The memories are carried per token: each of the $N = 100$ grid positions has its own recurrent slot, updated independently by a

shared LSTM cell, so spatial layout is preserved across time. The update is

$$H_1^{(t)} \leftarrow \text{LSTM}_1(X^{(t)}, H_1^{(t-1)}), \quad H_2^{(t)} \leftarrow \text{LSTM}_2(Z_{\text{pri}}^{(t)}, H_2^{(t-1)}),$$

so the fast memory H_1 is written directly from the image while the deep memory H_2 is written from the priority output and is therefore the channel through which the priority stream’s read propagates forward to the value stream on the next frame. This shared deep memory is the only forward coupling between the streams; there is no path by which the value stream writes anything the priority stream reads.

The read-out is split: the actor reads only Z_{pri} and the distributional critic reads only Z_{val} . Each head arranges its stream as (B, d, N) and applies three strided 1-D convolutions over the token (spatial) axis followed by a linear map — the actor to two wait/press logits, the critic to 51 quantiles per action. Reading over the token axis with a convolution preserves the spatial arrangement of the percept rather than pooling it away. The full model has 1,015,656 parameters.

Training. The agent is trained end-to-end by reinforcement with no auxiliary perceptual loss. The actor uses a policy-gradient objective in the maximum-a-posteriori policy-optimisation family — a closed-form weighted maximum-likelihood step toward actions the critic values, blended with a behaviour-cloning term — rather than a clipped surrogate; the critic is distributional, predicting 51 quantiles of the return per action and trained by a quantile-Huber loss against a periodically-refreshed target network. Episodes are drawn from a prioritised replay buffer (priority proportional to the per-step value error, with importance-sampling correction), and a short burn-in at the start of training forces the wait action so the critic and the recurrent representation warm up on full-length trajectories before the policy begins to act. The discount is 0.99. No part of the training signal specifies where to attend, what to ignore, or when to press; the spatial filter, the value gate, and the division of labour analysed below are all emergent from reward maximisation under this architecture.

2.3 Faithful attention extraction and intervention

Every mechanistic analysis is built on a hand-written reconstruction of the dual-stream step that recomputes each stream’s single-head cross-attention from the network’s own projection weights, returns the full attention tensors (priority $(B, 1, N, 2N)$, value $(B, 1, N, N)$), and allows an additive bias to be injected into either stream’s pre-softmax logits before the softmax. In evaluation mode this reconstruction reproduces the stock network’s two stream outputs, both memories, and — through the split read-out — the actor logits and critic quantiles to a maximum absolute deviation below 10^{-6} at zero bias, so the causal manipulations act on the real circuit. Because the image queries memory, the spatially-meaningful quantity is the weight each memory position receives, averaged over the image queries and laid on the 10×10 grid; for the priority stream, whose keys are $[H_1 \| H_2]$, we read the spatial map from its H_2 key block so both streams align to the four Gabor quadrants. For the attention-map and causal analyses we use forced trials — the cue side, target location, distractor location, and the change and distractor times held fixed while the random orientations average out — so that maps and perturbations are interpretable and matched across conditions.

2.4 Analyses

All analyses use a single frozen snapshot of the trained agent and the faithful reconstruction of Section 2.3.

Behaviour. We run greedy (argmax) roll-outs on randomly-sampled trials (3,000) and score each as a hit (press at or after the target change on a target trial), a miss (no press on a target trial), an early false alarm (press before the target change), or a correct rejection (no press on a no-target trial). We break these down by cue colour, displayed reliability, the target’s quadrant (primary versus secondary), and the presence and onset time of the distractor. Reaction time is the press frame minus the change frame on hits. Signal-detection sensitivity d' and criterion are computed from the hit and false-alarm rates within the engaged (red, green) regime.

Psychophysics. To fit psychometric and chronometric functions we use forced trials (Section 2.3) with the cue side, distractor, and timing fixed and the target’s orientation-change magnitude swept over $[3^\circ, 64^\circ]$, separately for valid (target at the cued primary quadrant) and invalid (secondary quadrant) targets at each displayed reliability. We fit a 4-parameter logistic, $P(\text{hit}) = \gamma + (1 - \gamma - \lambda) \sigma((m - \alpha)/\beta)$, with guess γ , lapse λ , threshold α , and slope β , by bounded least squares; the threshold α is the magnitude at the function’s midpoint.

Attention maps. On forced trials grouped by condition (cue side, cue colour, distractor present/absent) we average the full per-key attention — meaned over heads and image queries — at each timestep and lay it on the 10×10 grid, reading the priority stream from its H_2 key block. Because the change and distractor times are fixed in forced trials, timesteps align across trials and the averaged maps are clean. Per-quadrant time-courses sum the map over each Gabor quadrant’s 25 tokens.

Decoding. From per-quadrant latents (each stream and memory pooled within each quadrant) we train ℓ_2 -regularised logistic classifiers with standardised features and report balanced accuracy under stratified cross-validation, against a shuffled-label baseline. We decode the target location and distractor location (four-way), distractor presence and cue side (binary), and cue colour (three-way), both at a fixed probe two frames after the change and, with the change and distractor times fixed, at every timestep.

Causal perturbation. On matched, seeded forced trials we inject an additive bias into the pre-softmax attention logits of one stream’s keys for one region — the cued primary quadrant, the cued secondary quadrant, or the two distractor quadrants — and sweep it from -6 to $+6$ logits. We read the decision (hit and false-alarm rate, greedy policy) and the critic (state value and outcome uncertainty at the pre-change frame, where all trials are still live), and compute the d' shift.

Value. Under a forced-wait policy we track state value and the press-versus-wait advantage through the trial by cue colour, and the critic’s outcome uncertainty against change magnitude. Calibration compares the critic’s predicted value at each visited state to the realised discounted Monte-Carlo return under the greedy policy.

2.5 The architecture is not arbitrary

The two design features that matter for the analysis below — the shared recurrent memory across the streams and the routing of the grounded priority read to the actor — are both necessary for the agent to learn at all. In companion ablations on the distractor-free task, cutting the shared memory (two independent recurrent modules, each reading and writing only its own state) collapses the agent to an always-wait policy that never detects a change, as does routing the memory-gated value stream to the actor in place of the priority stream; only the shared-memory, priority-to-actor configuration studied here reaches competent detection. The shared deep memory is what keeps the critic’s valuation grounded in the same evolving state the actor acts from — a precondition for the credit assignment that the reinforcement learner needs — and the grounded priority read is

what gives the policy a stimulus-locked decision variable. The agent dissected here is that working configuration, taken intact and stress-tested under distraction and reward pressure.

3. Results

All results are from a single frozen snapshot of the trained agent. We first establish what it does — a value-gated, spatially-selective strategy whose aggregate accuracy (0.60) hides a sharp internal structure — then take the circuit apart to show how and where each selectivity is computed.

3.1 Behaviour: a value gate over a near-perfect detector

Scored over all trials the agent looks middling: 60% correct, a 0.54 hit rate, and a high 0.36 early-false-alarm rate on target trials. This aggregate is misleading. Broken down by the reward at stake, the agent splits trials into two regimes (Figure 1A). On the two high-value cue colours it is an excellent detector: it catches 82.5% of red ($v = 5$) and 83.7% of green ($v = 3$) target changes while almost never false-alarms (early-false-alarm rates 0.007 and 0.002). On the low-value colour it abandons the task entirely: it detects 0% of blue ($v = 1$) changes, and instead presses at the second frame on essentially every blue trial — a deliberate bail that ends the low-value episode at once. The whole 0.36 aggregate false-alarm rate is this blue bail; on the trials the agent chooses to engage, it false-alarms less than once in a hundred. Signal-detection analysis confirms a sharp engaged detector with a conservative criterion: $d' = 3.41$ (red) and 3.86 (green), criterion +0.77 and +0.95.

The agent retains a spatial-cueing signature within the engaged regime (Figure 1C): comparing the cued (primary) quadrant against the same-side secondary quadrant, the cued location is detected slightly more often at every reliability (benefit +0.02 to +0.04). Responses are fast and change-locked (median reaction time 1.6 frames after change; Figure 1D).

The headline of the behavioural analysis is what does not drive the errors (Figure 1B). The distractor — the uncued, equally-salient change the agent is trained to ignore — has essentially no behavioural effect. On engaged trials the early-false-alarm rate is 0.002 when no distractor is present and 0.000 when a distractor appears before the target; the hit rate is 0.820 versus 0.808. And of every early false alarm the agent makes on a target trial, none occurs at or after a distractor onset — the premature presses are early impulsive bails (median press time $t = 2$, before the earliest possible distractor at $t = 11$), not distractor captures. The distractor is, behaviourally, a non-event; the agent’s “struggle” is entirely the blue abandonment.

3.2 Psychometric and chronometric functions

Detection is a graded function of the physical signal (Figure 2A). Fitting 4-parameter logistic functions to hit rate versus orientation-change magnitude for engaged trials, sensitivity is high — the hit rate climbs from near zero to ceiling over roughly 5–20° — and reaction time falls correspondingly from seven to ten frames near threshold to one or two frames for large changes (Figure 2B). The cueing signature appears as a leftward shift of the psychometric function for valid targets: at every displayed reliability the valid-target detection threshold is lower than the invalid one ($\approx 7.5^\circ$ versus $\approx 10^\circ$; Figure 2C), so a validly-cued change is detected at a smaller magnitude. The benefit measured at a fixed near-threshold magnitude is modest (+0.02 to +0.07 at 16°, largest at full reliability) and does not scale cleanly with reliability the way it does on the distractor-free task — under distraction the agent holds a fixed, target-locked filter (Section 3.3) rather than finely

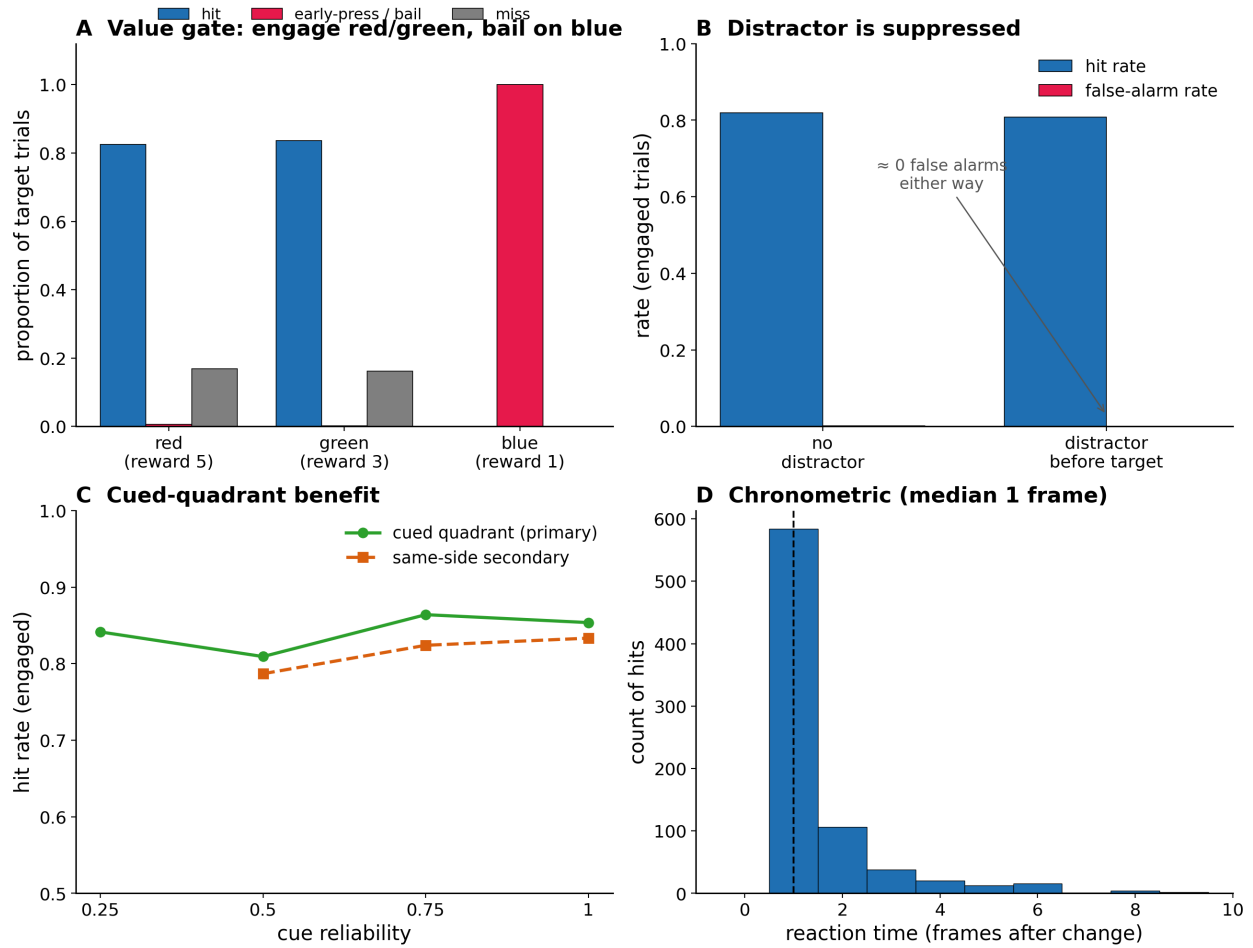


Figure 1: Behaviour. A Outcome of target trials by cue colour: a near-perfect detector on high-value red and green (high hit, near-zero false alarm) that abandons low-value blue, bailing with an immediate press. B Distractor suppression on engaged trials: the false-alarm rate stays at zero whether or not a distractor precedes the target, and the hit rate is essentially unchanged. C Within the engaged regime the cued (primary) quadrant retains a small detection benefit over the same-side secondary quadrant. D Reaction times are fast and change-locked (median 1.6 frames).

re-weighting it by cue reliability. Sensitivity is the robust cueing signal here; the fixed-magnitude benefit is compressed because engaged detection is near ceiling for all but the smallest changes.

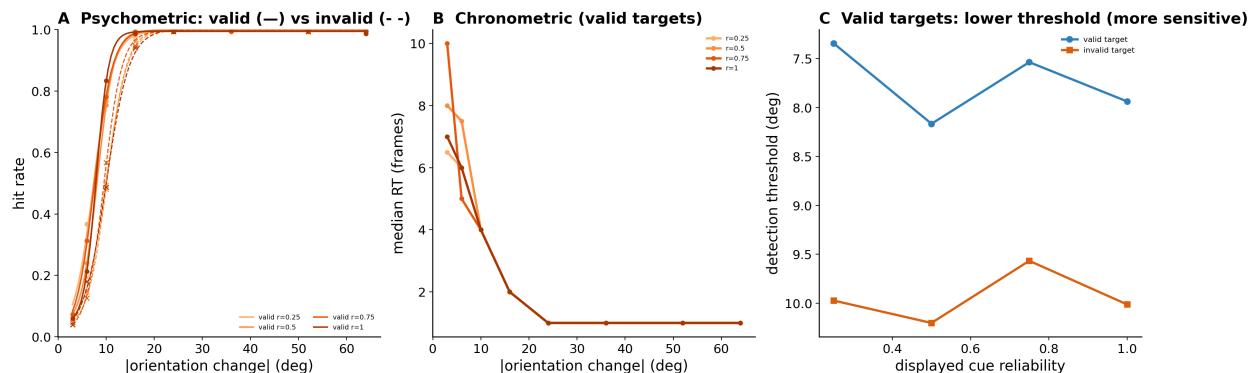


Figure 2: Psychometric and chronometric functions (engaged trials). A Hit rate versus change magnitude with 4-parameter logistic fits, valid (solid) and invalid (dashed) targets at each displayed reliability. B Reaction time falls with magnitude. C Valid targets have a lower detection threshold than invalid targets at every reliability (axis inverted so higher = more sensitive) — the cueing signature as a sensitivity shift.

3.3 The priority stream filters the distractor

Why is the distractor a behavioural non-event? Because the stream that drives the policy does not read it. Averaging the priority stream’s full 10×10 attention map over forced trials (Figure 3) shows attention concentrated on the cued side and, within it, sharpening onto the cued primary quadrant as the trial unfolds; the distractor quadrant stays dark throughout, even after the distractor changes. The pattern is cue-driven, not a fixed spatial habit: it mirrors when the cue points right (row 2), placing attention on the opposite side. On no-distractor trials (row 3) the map is the same minus the (already cold) distractor.

Quantified per quadrant (Figure 4A), priority attention to the cued primary quadrant rises anticipatorily from 0.17 just after the cue to 0.33 before the change and holds at 0.26 through and after it, while the distractor quadrant falls from an already-low 0.042 to 0.004 and the cued secondary and far quadrants are driven to near zero. The cued-versus-distractor selectivity at the change is $\approx +0.21$. Crucially the suppression is proactive, not a reaction to the distractor once it appears: aligning distractor-quadrant attention to the distractor’s own onset, a distractor change raises it by only +0.004 (from 0.042 to 0.045). An equally salient orientation step that on the cued side would draw the attention that triggers a press draws essentially nothing when it falls in the region the priority stream has learned to leave unread.

3.4 The value stream tracks the scene — including the distractor

The distractor is not, however, globally suppressed: the agent is not blind to it. The value stream — which feeds the critic — behaves quite differently from the priority stream (Figure 5). Its read is broader and less sharply target-locked, and, tellingly, it does attend the distractor quadrant: when the distractor changes (column 2 onward) a bright spot appears on the distractor side of the value map, absent from the priority map at the same moment, and the per-quadrant time-course (Figure 4B) shows a transient rise in value distractor-quadrant attention that the priority stream does not have. The distractor change is therefore registered by the agent — but only in the stream that

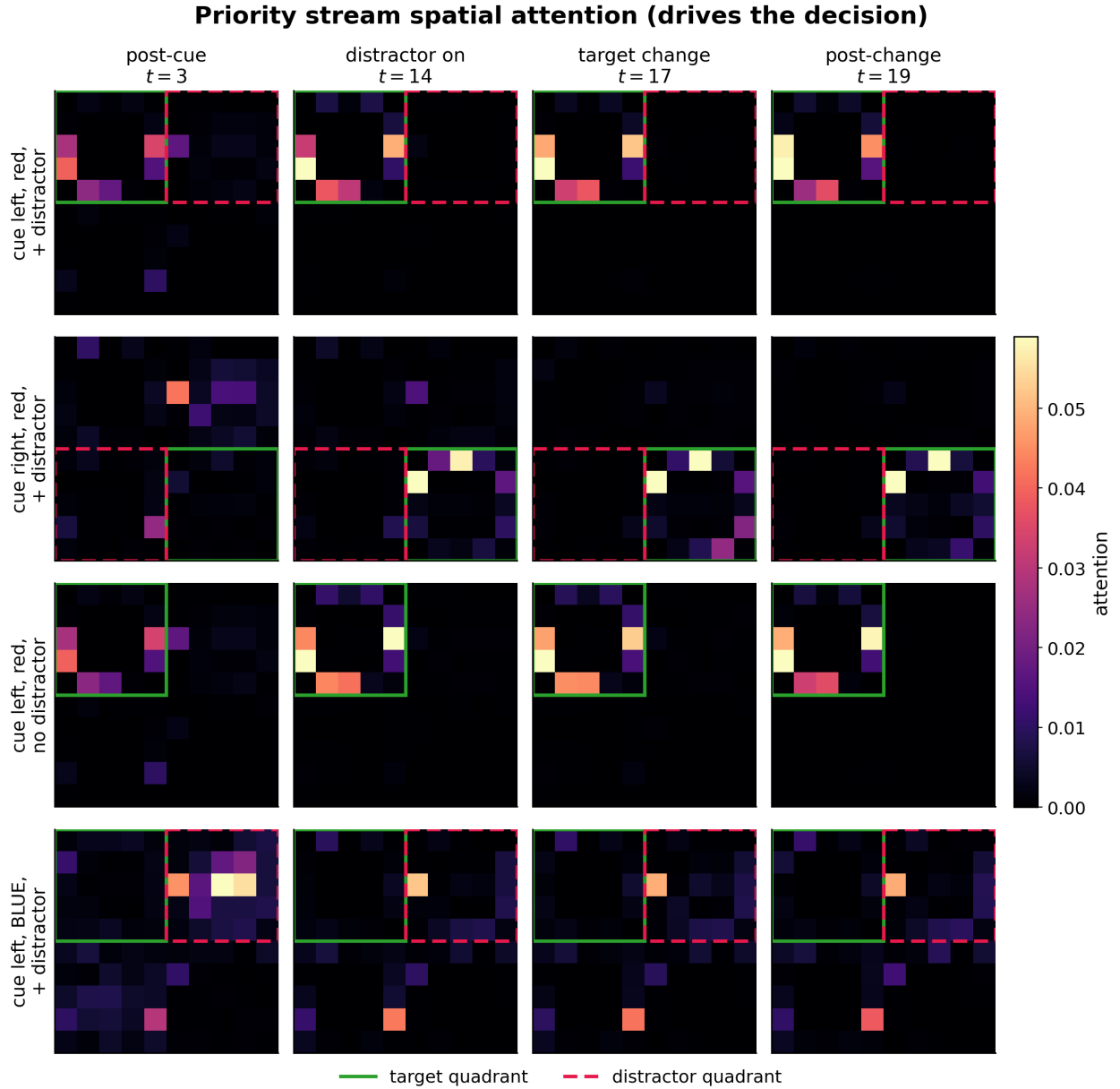


Figure 3: Priority-stream spatial attention (drives the policy), averaged over forced trials. Rows: cue-left red with a distractor, cue-right red with a distractor (the mirror), cue-left red with no distractor, and cue-left blue with a distractor. Columns: four timepoints from post-cue to post-change. Green box = target quadrant, dashed pink = distractor quadrant. The priority read concentrates on the cued side and sharpens onto the target quadrant while keeping the distractor quadrant cold — except on the low-value blue trial (row 4), where it is diffuse and leaks onto the distractor (Section 3.5).

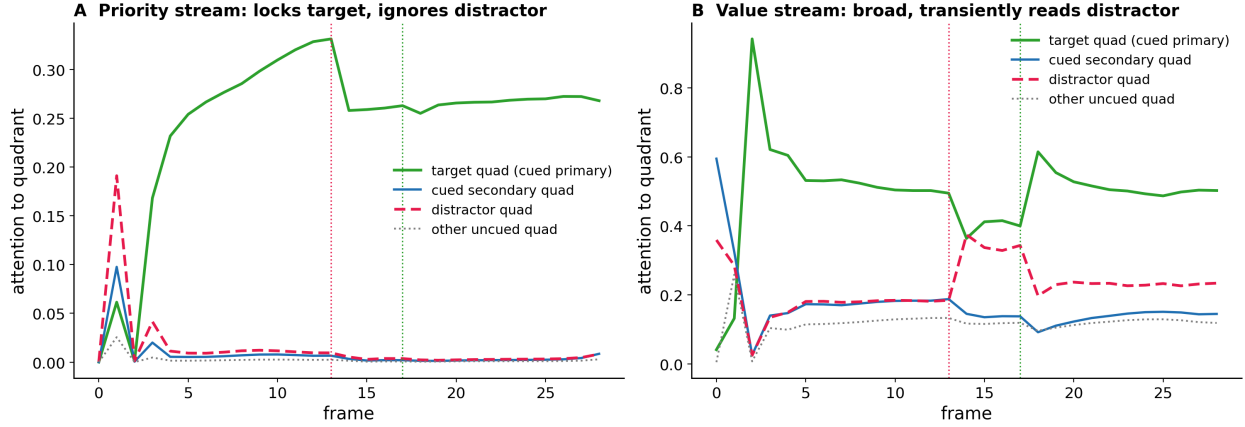


Figure 4: Per-quadrant attention time-courses (forced cue-left, red, with distractor). A The priority stream locks onto the target quadrant (rising anticipatorily, holding through the change) and drives the distractor and other quadrants toward zero. B The value stream holds a broader, cue-anchored read of the cued side and, unlike the priority stream, transiently reads the distractor quadrant when it changes (Section 3.4).

sets value, not the stream that sets the decision. This is why the competing change can inform the critic’s estimate of the scene while never reaching the response: the filter that protects behaviour lives specifically in the decision pathway, and the value pathway is free to see everything. Sections 3.6 and 3.7 confirm this representationally and causally.

3.5 Value reshapes the filter

The spatial filter is not a fixed piece of wiring; the agent deploys it in proportion to what the trial is worth. Holding the stimulus, cue side, and target and distractor locations fixed and varying only the cue colour, the priority stream’s read is sharply different for a high-value and a low-value trial (Figure 6, and Figure 3 row 1 versus row 4). On a high-value (red) trial the priority filter is tight and target-locked: attention to the target quadrant peaks at 0.33 and the distractor quadrant is held at 0.005. On a low-value (blue) trial the same stream, seeing the same image, is diffuse and poorly targeted: attention to the target quadrant reaches only 0.10, and — the telling part — it leaks onto the distractor, which now draws 0.13, thirty times more than on the red trial. The agent does not merely decide whether to act on a low-value trial; it does not bother to sharpen the perceptual filter that would suppress the distractor, because on a trial it intends to abandon there is nothing to protect. Distractor suppression is thus itself value-gated — near-perfect where detection pays, barely deployed where it does not. (This corrects a natural but wrong reading on which the value gate is purely post-perceptual: value reaches into the attention front-end and tunes the filter’s sharpness, not only the downstream choice.)

Repeating the matched comparison across all three reward levels shows the modulation is not a graded gain but is tied to the binary engage/abandon decision (Figure 7). The two engaged colours are indistinguishable — red and green both hold target-quadrant attention at 0.26 and the distractor at 0.003–0.004 — while only the abandoned blue collapses (target 0.034, distractor 0.111). The agent does not turn its filter up a little more for red than for green; it deploys the sharp, distractor-proof filter on any trial it intends to engage and lets it lapse on the trial it intends to abandon. The spatial filter and the value gate are thus locked together: the same decision that determines whether to act determines whether to perceive selectively.

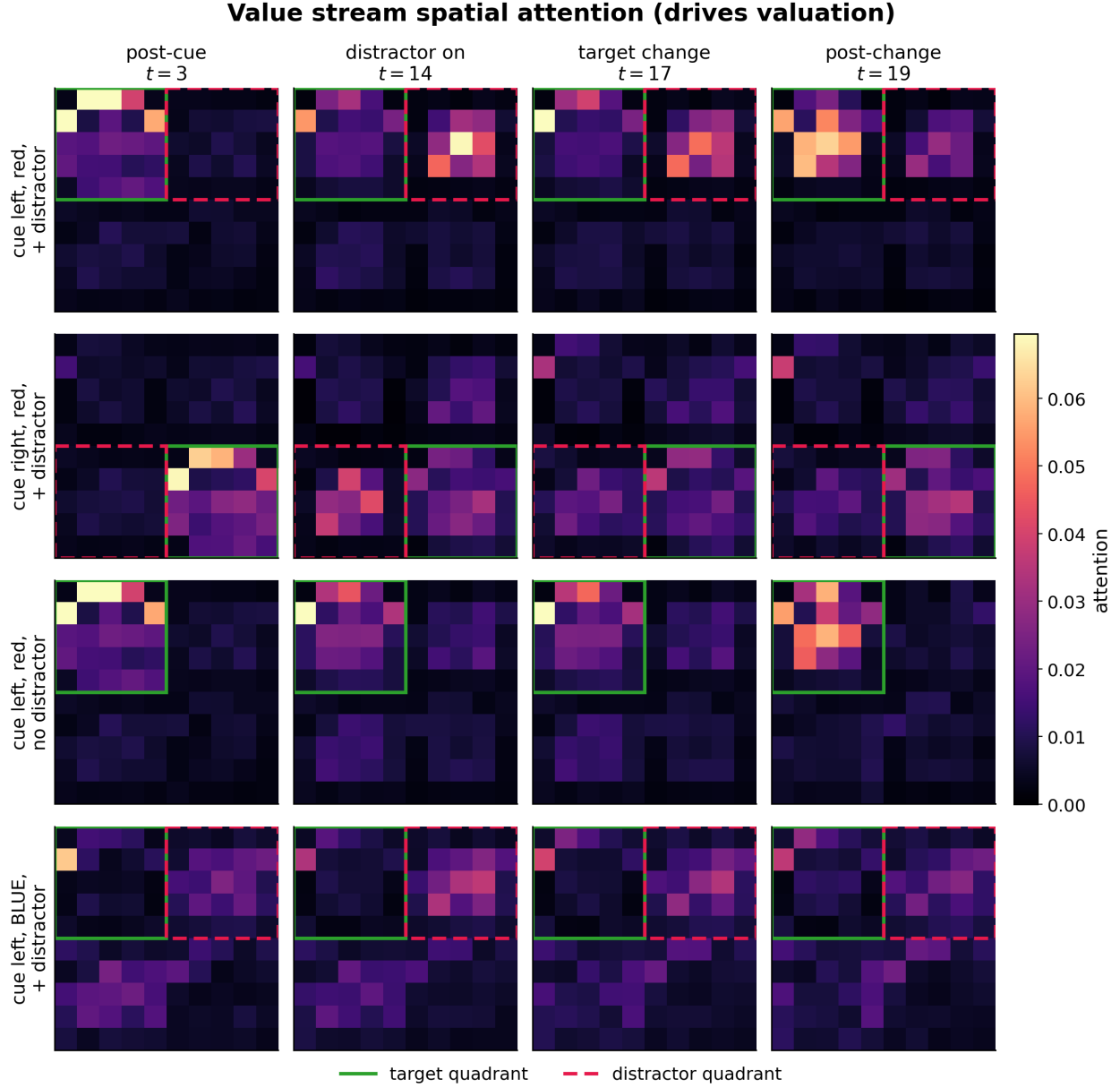


Figure 5: Value-stream spatial attention (drives value), same layout as Figure 3. The value read is broader and cue-anchored, and — unlike the priority stream — it transiently attends the distractor quadrant when the distractor changes (column 2 onward), so the critic registers the competing change even though the actor’s stream filters it out.

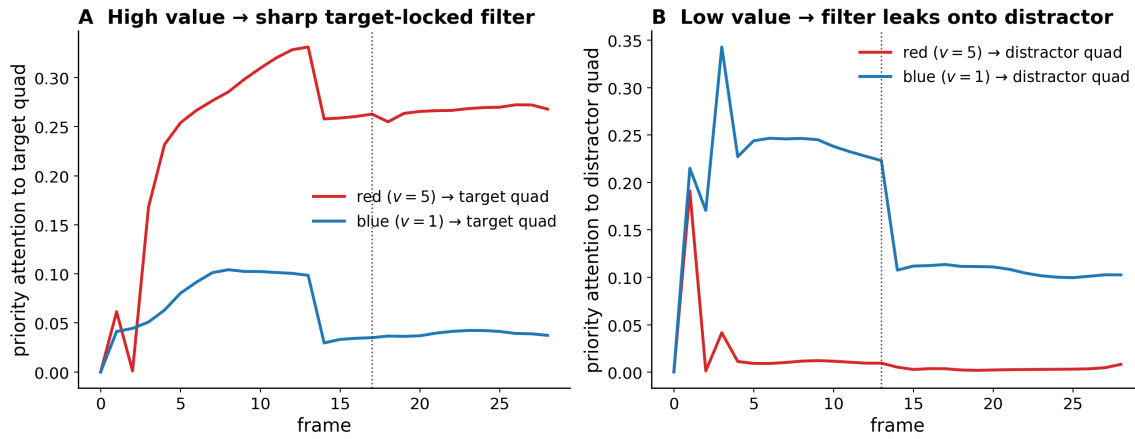


Figure 6: Value reshapes the spatial filter (matched forced trials, only the cue colour differs). A Priority attention to the target quadrant is high and anticipatory for a high-value (red) cue and weak for a low-value (blue) cue. B Priority attention to the distractor quadrant stays near zero for red but rises substantially for blue — the filter is sharp where the trial pays and leaky where it does not.

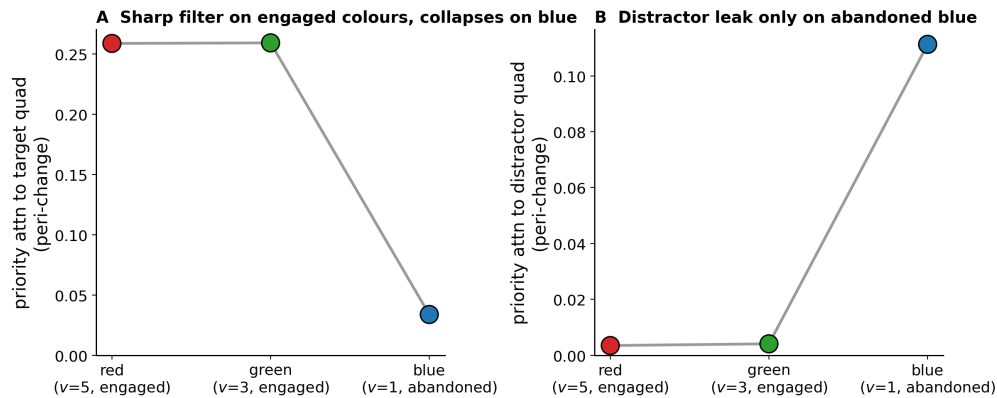


Figure 7: The filter is engage-gated, not reward-graded. Priority attention to the target quadrant (A) and the distractor quadrant (B) peri-change, for the three cue colours. The two engaged colours (red, green) are indistinguishable — both sharp on the target and cold on the distractor — and only the abandoned colour (blue) shows a collapsed, distractor-leaky filter.

3.6 What each stream represents, and when

Linear decoding confirms and localises the picture. At a fixed probe two frames after the change (Figure 7B), the target location is decodable at high accuracy from every representation, including the actor’s priority stream (0.89, four-way, chance 0.25). The distractor location tells the filtering story: it is fully present in the raw sensory memory H_1 (0.94, identical to the target), but markedly less recoverable from the actor’s priority read (0.77), and the mere presence of a distractor is decodable least of all from the priority stream (0.69, the lowest of the four representations) while being stronger in the critic’s stream (0.80). The distractor is in the input but selectively de-emphasised in the readout that drives the policy. Cue colour (value) and side decode at ceiling everywhere, including the priority stream (0.97, 0.89), so the value gate has all the information it needs in the policy’s own input.

Decoding at every timestep (Figure 7A) shows the dynamics. Cue colour and side are decodable at ceiling from cue onset and held throughout. The target location comes online sharply after the change, rising in the priority stream to ≈ 0.85 . The distractor’s presence and location rise only after the distractor onset and to a much lower level in the priority stream — and, resolving the two streams (Figure 7C panel), the distractor location is consistently better decoded from the critic’s value stream than from the actor’s priority stream, the temporal counterpart of the spatial-map result in Section 3.4. The filter is not a momentary suppression but a sustained representational gap between target and distractor that is specific to the decision pathway.

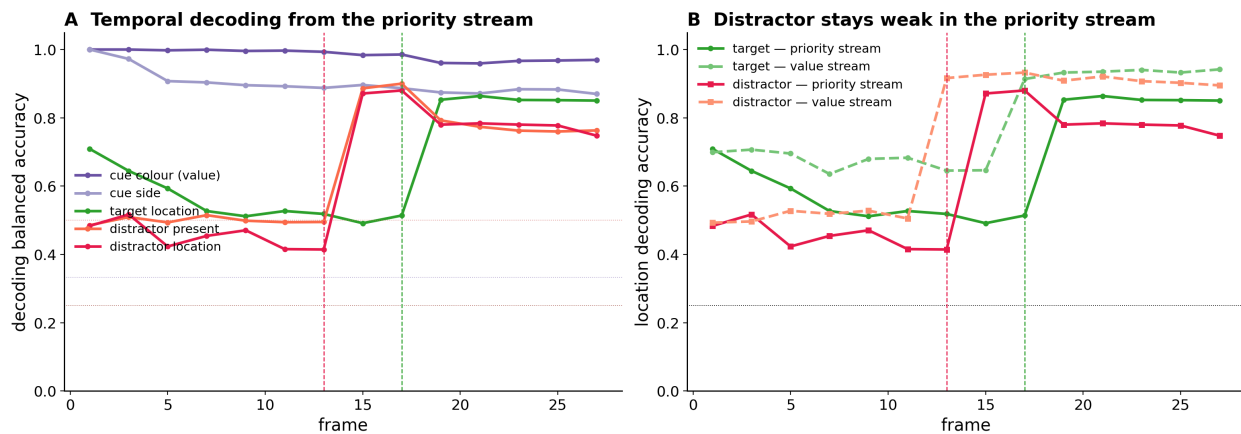


Figure 8: Representational decoding. Left Temporal decoding from the actor’s priority stream: cue colour and side at ceiling from onset; target location rises sharply after the change; distractor presence and location rise after the distractor onset but stay weak. Right Target versus distractor location decoded from the actor (priority) and critic (value) streams — the distractor is better decoded from the critic stream, the representational signature of a decision-pathway-specific filter.

3.7 Causal dissociation: priority moves the decision, value moves value

The architecture predicts, and the perturbations confirm, a clean causal division of labour (Figure 8). On forced engaged trials we inject an additive bias into one stream’s read of one quadrant’s keys and sweep it from -6 to $+6$ logits, reading both the decision and the critic.

Priority attention controls the decision. Biasing the priority read of the cued primary quadrant swings the hit rate across its entire range — from 0.01 at the strongest suppression to 1.00 at the strongest enhancement — and the false-alarm rate inversely, while the critic’s value barely

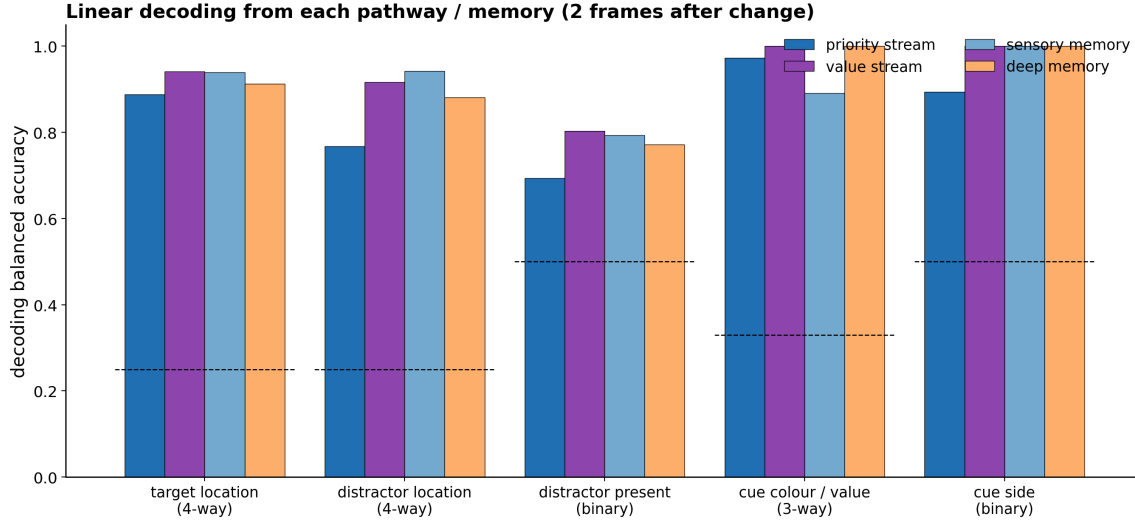


Figure 9: Single-timepoint decoding two frames after the change, all four representations. The target location decodes from every stream including the actor’s; the distractor location and presence decode worst from the actor’s priority stream though the raw sensory memory holds them fully; cue colour (value) and side decode at ceiling.

moves (3.93 $\rightarrow$ 3.98). Suppressing the cued quad’s priority read abolishes detection (hit 0.01): the agent relies on attending the cued location to detect the change. Biasing the priority read of the distractor side does the opposite and is the protective filter made causal — it drives the hit rate down (1.00 $\rightarrow$ 0.01) and manufactures false alarms (0.00 $\rightarrow$ 0.99), collapsing and then inverting sensitivity (d' 5.53 $\rightarrow$ 2.91 $\rightarrow$ -4.90). Forcing the agent to read the distractor makes it respond to the distractor; the suppression is something the agent actively does, sited in where the priority stream reads, and removable by intervening there.

Value attention never moves the decision. Biasing the value read of either the cued primary quadrant or the distractor side leaves the hit and false-alarm rates flat (1.00 and 0.00 throughout) while shifting the critic’s value and outcome uncertainty. The stream that feeds the critic controls the valuation and its uncertainty, not the press. The dissociation is therefore not merely correlational but causal and complete: priority $\rightarrow$ decision, value $\rightarrow$ value, each stream commanding only the head it feeds. Combined with Section 3.4 — the value stream reads the distractor, the priority stream does not — this explains in mechanism why the competing change informs value but can never trigger a response.

3.8 A value-ordered, calibrated critic that gates engagement

The value gate of Section 3.1 is driven by the critic (Figure 9). Under a forced-wait policy the state value at the change rises in precise proportion to the reward at stake — 4.37 for red ($v = 5$), 2.57 for green ($v = 3$) and 0.21 for blue ($v = 1$) — a clean preservation of the 5:3:1 order, with blue valued roughly twenty times below red. Blue’s value is close to zero, and correspondingly the press-versus-wait advantage for blue hovers near zero throughout while it carries real magnitude for red and green; with nothing to be gained, the policy on a blue trial has no gradient pulling it toward careful detection and settles into the cheapest terminating action, the immediate bail. The critic is well-behaved otherwise: its predictions are well-calibrated against realised Monte-Carlo returns (slope 1.01, $R^2 = 0.41$ over $\sim 21,000$ states), and its distributional spread narrows as

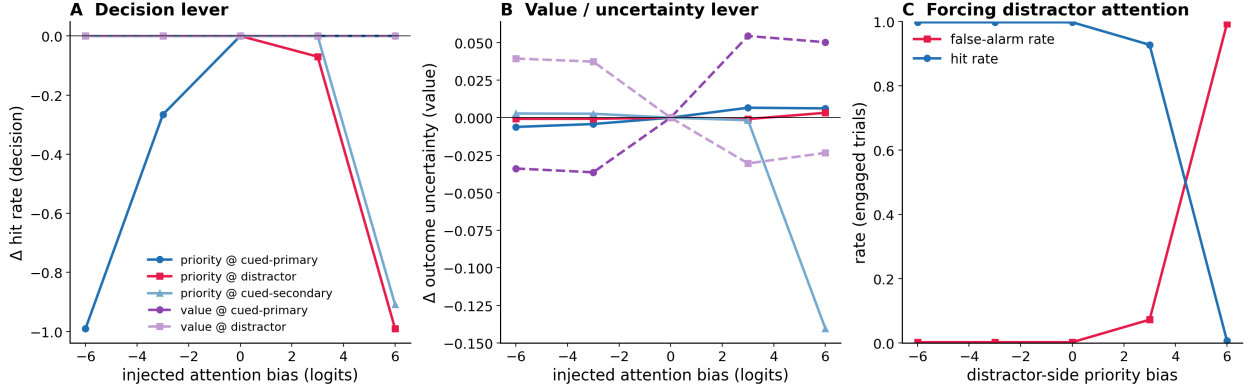


Figure 10: Causal perturbation battery (forced engaged trials, bias swept -6 to $+6$). A Effect on the decision (Δ hit rate): priority levers (solid) swing it from 0 to 1; value levers (dashed) are flat. B Effect on the critic's outcome uncertainty: value levers move it, the decision-controlling priority-primary lever does not. C Forcing distractor-side priority attention drives the hit rate down and the false-alarm rate up — the protective filter, made causal.

evidence accumulates (quantile standard deviation just after the change falling from 0.81 for the smallest changes to 0.23 for the largest). On the engaged regime the agent's sensitivity is high and its criterion conservative ($d' = 3.4\text{--}3.9$, criterion $+0.8\text{--}0.9$); when it chooses to play, it plays well and cautiously, which is why its engaged false-alarm rate is near zero even under distraction.

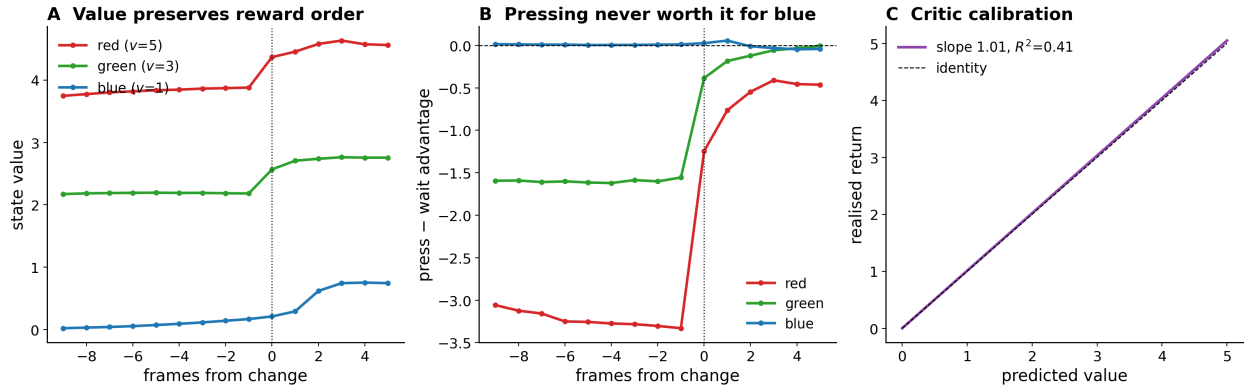


Figure 11: The critic. A State value by cue colour rises through the trial in proportion to the reward at stake (red $>$ green $>$ blue), with blue near zero. B The press-versus-wait advantage carries magnitude for red and green but stays near zero for blue — pressing on a low-value trial is judged not worth it, the driver of the bail. C The critic is well-calibrated against realised returns (slope 1.01).

4. Discussion

4.1 A causal division of labour between policy and value

The clearest structural fact about this agent is that its two streams command two different heads, and the perturbations make this causal rather than merely architectural. Biasing the priority stream moves the decision across its whole range and leaves value untouched; biasing the value stream moves value and uncertainty and never touches the decision. The policy and the valuation are separable computations, each controlled by the stream that feeds it, sharing only the recurrent

memory through which the priority output is written forward. This separation is what makes everything else in the agent possible: it lets the same competing change be handled differently by the two systems, and it lets effort be priced by one system and spent by the other.

4.2 Distractor suppression lives in the decision pathway, not in perception writ large

The agent's robustness to the distractor is often summarised as "it ignores the distractor," but the mechanism is more specific and more interesting. The agent is not blind to the distractor: the raw sensory memory encodes it fully, and the value stream actively reads it when it changes, so the critic registers the competing event. What the agent does is withhold the distractor from the decision pathway — the priority stream keeps the distractor quadrant cold, the distractor decodes worst from the priority read, and forcing priority attention back onto the distractor manufactures the false alarms the filter prevents. Suppression is thus a property of one pathway, not of perception as a whole; the agent sees the distractor and declines to act on it, rather than failing to see it. This is a sharper claim than global filtering and a more plausible model of how a system can both monitor its surround for value and refuse to be driven by it.

4.3 Value reaches into perception: the filter is tuned by what is at stake

A natural first reading — that the value gate is a purely downstream decision, leaving perception untouched — is wrong, and the matched-stimulus attention maps are what overturn it. With the image held fixed and only the cue colour changed, the priority filter is sharp and distractor-proof on a high-value trial and diffuse and distractor-leaky on a low-value one. Value does not merely decide whether to act; it sets whether the perceptual filter that suppresses the distractor is deployed at all. The modulation, moreover, is not a graded gain tracking the exact reward — the two engaged colours are perceptually identical — but a binary switch tied to the engage/abandon decision: the filter is sharp on any trial the agent will play and collapses only on the one it will abandon. This couples the two selectivities that we otherwise describe separately: the value gate and the spatial filter are not independent modules but a single policy in which one decision — is this trial worth it — controls both whether to engage and whether to perceive selectively. It connects to the proposal, in the attention literature, that attentional deployment follows the value of a location, but sharpens it: here the coupling is all-or-none and learned, from reward alone, as part of the same engage decision rather than as a separate gain control.

4.4 Reading the aggregate correctly, and the value gate

A single accuracy number badly mischaracterises this agent. At 0.60 correct it reads as a mediocre detector struggling with a hard task. The decomposition shows the opposite: on the two-thirds of trials it engages it is a near-ceiling detector (0.83 hit, < 0.01 false alarm, $d' \approx 3.7$) that is almost perfectly immune to the distractor, and the missing third is not failure but refusal — a deliberate abandonment of trials a calibrated critic prices at essentially zero. Notably, on the distractor-free version of this task the same architecture does not abandon low-value trials; under distraction it has learned to gate hard on value, plausibly because engaging now carries the cost of holding a guarded, target-locked, distractor-suppressing filter that is not worth deploying for a reward of one. The value gate is the rational endpoint of pricing effort against stake, and the agent's modest aggregate is the visible cost of that pricing, not a limit on its competence. The general caution for evaluating learned agents follows: aggregate performance on heterogeneous-value tasks conflates competence with willingness, and the structure lives in the breakdown.

4.5 Cueing under distraction

The spatial-cueing signature survives the addition of a distractor but in a muted, robust form: valid targets are detected at a lower threshold than invalid ones at every reliability, while the fixed-magnitude validity benefit is small and does not scale cleanly with displayed reliability. The attention maps explain why — the agent holds a fixed, strongly target-locked filter rather than re-weighting it finely with the cue’s stated reliability. Under distraction the premium is on a stable filter that reliably excludes the uncued side, and the agent appears to trade the fine reliability-scaling it shows on the distractor-free task for that stability. The benefit that remains is carried as sensitivity (a threshold shift) rather than as a near-ceiling accuracy difference.

4.6 Relation to theories of attention: biased competition and the priority map

The mechanism we describe maps closely onto the biased-competition account of attention, in which representations of different locations compete for control of behaviour and top-down signals bias that competition toward task-relevant ones. The agent’s priority stream is, functionally, a priority map: it assigns a weight to each location, that weight competes (through the softmax) for the read that drives the policy, and the cue biases it toward the cued side in advance of any stimulus. What the distractor task adds is a clean test of the resolution of that competition. A salient distractor is exactly a strong bottom-up competitor; in a system that merely pooled saliency it would win attention and drive a response. Here it does not, because the bias the cue installs is strong enough that the distractor never enters the competition that the policy reads — while still entering the broader read that the value system maintains. The dissociation between the two streams is therefore a concrete instance of a priority map that is consulted differently by the action and the evaluation systems: a single scene, two competitions resolved differently, one feeding choice and one feeding worth. The persistence of a cued-side bias before the change (the anticipatory rise in priority attention to the cued quadrant) is the model’s version of the spatial pre-allocation that defines covert attention.

This also connects to the “attentional template” idea, in which a held representation of the task-relevant location or feature biases ongoing perception. The cue’s side and reliability are written into the recurrent memory at cue time and decode from it at ceiling throughout the trial; that stored template is what the priority stream reads against, and it is what keeps the filter pointed at the cued side across the seconds between cue and change. The template here is learned and value-weighted: it carries not only where to look but, through the engage gate, whether to bother looking sharply at all.

4.7 Reward, value, and the control of attention

The agent learns to allocate attention purely in the service of reward, and the result is a system in which value controls attention at two levels. At the level of engagement, a calibrated critic prices each trial and the policy declines the cheap ones — and, strikingly, declines to deploy the spatial filter on them, so that perception itself becomes contingent on worth. At the level of allocation within an engaged trial, the cued-side bias and the distractor suppression are held in place by the same reward pressure that makes detection worthwhile. This is consistent with proposals that attention is drawn toward valuable locations and that reward shapes attentional priority, but it sharpens the prediction in a specific way: the value control we observe is not a graded gain proportional to expected reward but a binary gate that couples to the decision to engage. An agent built to maximise reward, with separate read-outs for policy and value, discovers that the

economical solution is to switch the expensive, distractor-suppressing filter on and off with the engage decision rather than to modulate it continuously. Whether biological value-based attention is graded or similarly gated is an empirical question this model frames concretely.

The division of labour we find — a decision stream that filters the distractor and a value stream that still tracks it — also speaks to the relationship between attention and value estimation. Because the critic reads the stream that retains the distractor, the agent’s valuation of the scene can incorporate the competing event even though its choice cannot be driven by it. A system that needs to keep monitoring its surround for changes in value while refusing to be captured by salient irrelevancies has, in this architecture, a natural way to do both at once: route the broad, capture-prone read to the evaluator and the narrow, protected read to the actor.

4.8 Neural parallels and testable predictions

We are agnostic about anatomical correspondences, but the functional roles invite parallels. The priority stream’s causal control over the decision, and its spatial, cue-biased, change-locked attention, resemble the role attributed to fronto-parietal and collicular priority maps (FEF, LIP, SC) in covert orienting, where micro-stimulation or inactivation shifts behaviour at the attended location. Our perturbations are the in-silico analogue: biasing the priority read of a location shifts the decision exactly as enhancing or suppressing a priority-map node is expected to. The value stream’s control over value and uncertainty, with no effect on the choice, resembles the role of value-coding circuitry (orbitofrontal and ventromedial prefrontal cortex, and midbrain dopaminergic targets) that scales with expected reward and outcome uncertainty without itself issuing the motor command. The model makes two concrete, testable predictions in this frame. First, manipulating activity at an attended location should dissociably affect detection versus value-decodability depending on whether the manipulated population is in the decision or the evaluation pathway — enhancing the decision pathway should raise both detection and the decodability of the stimulus, while enhancing the evaluation pathway should move estimated value with little effect on the choice. Second, a salient uncued distractor should be more decodable from value-related populations than from choice-related ones even when it has no behavioural effect — the neural signature of seeing-but-not-acting that the two streams here display.

4.9 Limitations and outlook

The agent is analysed at a single training snapshot and seed; it is competent and stable but the causal effect sizes should be read as directional, even where, as here, they are large. The distractor is spatially separated from the target (opposite side) and the task is low-bandwidth (one change of one attribute in one location), which makes the decision-pathway filter legible but does not test harder cases — distractors adjacent to the target, multiple simultaneous distractors, or distractors sharing the target’s colour, any of which would stress a purely spatial filter. The filter’s engage-gating is established across the full reward range here (red and green identical, blue collapsed); whether a finer reward structure, or training to convergence, would ever convert the binary switch into a graded gain remains open, as does whether the gating is present early in training or emerges with the distractor demand. Our interventions act on attention logits, the most interpretable lever; the streams also interact through the shared deep memory, and perturbing that pathway directly would test whether the decision-pathway filter can be lifted from inside the recurrence as well as from the read. With those caveats, the account is concrete and mechanistic: two streams with a causally verified division of labour; a distractor filter that sits in the decision pathway while the value

pathway still tracks the scene; a filter whose sharpness is set by the trial's worth; and a calibrated critic that gates engagement.

5. Conclusion

A compact recurrent vision agent, trained only by reward to detect cued changes while ignoring uncued distractors, solves the problem with a causally verified division of labour and two interacting selectivities. Its priority stream carries the decision and its value stream carries the valuation, as direct perturbation shows: bias the first and the choice swings from never to always detecting; bias the second and only the value moves. On this substrate the agent filters the distractor in the decision pathway specifically — the priority stream holds the distractor cold while the value stream still reads it, so the competing change informs value but can never trigger a response, and forcing priority attention onto the distractor manufactures the false alarms the filter prevents. And it gates on value at both stages: a calibrated critic prices the reward order precisely and, on low-value trials, both relaxes the spatial filter (which goes diffuse and distractor-leaky) and abandons the trial outright. The agent's unremarkable aggregate accuracy is the sum of near-perfect, distractor-immune detection where it pays and a deliberate refusal to work where it does not. Where the agent looks solves the distractor, which stream looks decides whether what it sees can act, and what the trial is worth sets whether — and how sharply — it looks at all.

6. Supplementary analyses

6.1 Reaction-time and bail dynamics

The two response regimes are also cleanly separated in their timing (Figure 11). On engaged trials the hit reaction-time distribution is sharply peaked at one to two frames after the change — fast, change-locked detection. On abandoned (blue) trials the press-time distribution is a spike at the second frame: the bail is immediate and deterministic, occurring before the target-change window even opens (100% of blue presses fall before frame five, median frame two). The blue response is therefore not a degraded or mistimed detection but a categorical decision, taken at the start of the trial on the basis of the cue alone, not to engage. This is consistent with the value-gate account: the critic's near-zero valuation of blue is available from cue onset (cue colour decodes at ceiling from the first frames), and the policy acts on it immediately rather than waiting to see a change it has already decided is not worth detecting.

6.2 Anticipatory establishment of the filter

The spatial filter is set up by the cue and held across the cue-to-change interval, rather than being assembled when the change arrives. In the forced-trial maps the priority read is already biased to the cued primary quadrant by the third frame — well before the target change at frame seventeen — with target-quadrant attention at 0.17 just after the cue, climbing to 0.33 by frame twelve and holding through the change. Across the same interval the distractor quadrant is held low and is driven still lower as the trial proceeds. The cued-side advantage is thus a genuine pre-allocation, established from the cue and maintained over the delay, which is the model's counterpart of anticipatory covert orienting: the agent commits attention to the cued location in advance of any stimulus there, and that standing commitment is what makes the later change detectable and the contemporaneous distractor inert. The persistence of the bias over a multi-frame delay also implicates the

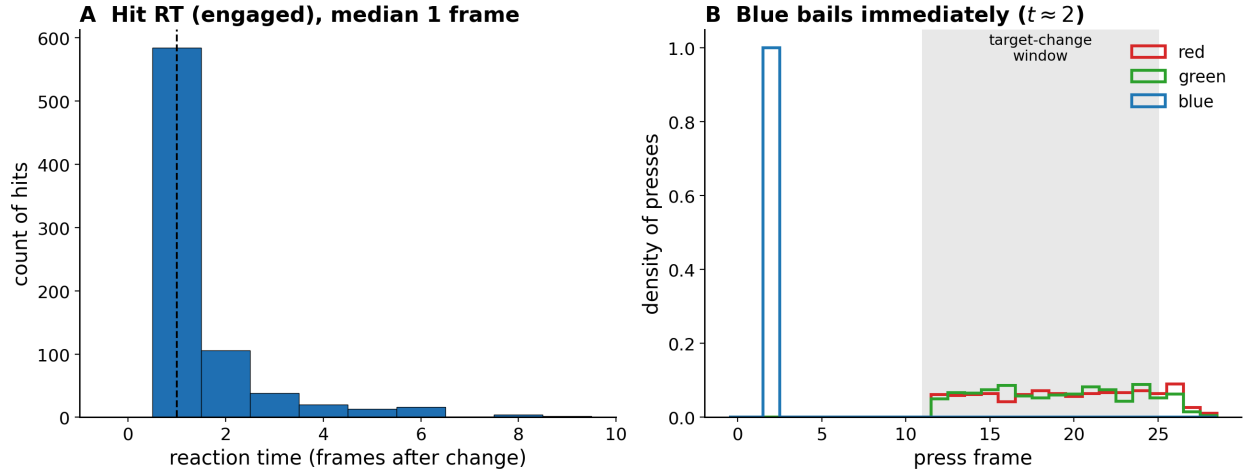


Figure 12: Supplementary: response timing. A Reaction-time distribution on engaged hits — fast and change-locked (median one frame after the change). B Press-time distribution by cue colour: red and green presses fall in the target-change window (shaded), whereas blue presses spike at the second frame, before the window opens — an immediate, categorical bail.

recurrent memory, into which the cue is written and from which the bias is continually read — the substrate by which a transient cue produces a sustained attentional state.

6.3 Robustness of the distractor-suppression result

The behavioural immunity to the distractor is not an artefact of a particular distractor timing or magnitude. It holds whether the distractor onsets before or after the target (engaged false-alarm rate 0.000 and 0.002 respectively), across the random distractor magnitudes sampled in the behavioural roll-outs, and — most tellingly — it is removable on demand: the causal sweep shows that a sufficiently strong bias onto the distractor side does produce distractor-driven false alarms, so the null effect under normal operation reflects an active filter at its working point rather than an insensitivity of the architecture. Conversely the dependence on attending the cued quadrant is equally real: suppressing the priority read of the cued quadrant abolishes detection. The agent’s distractor immunity and its target sensitivity are two faces of the same controllable spatial filter.